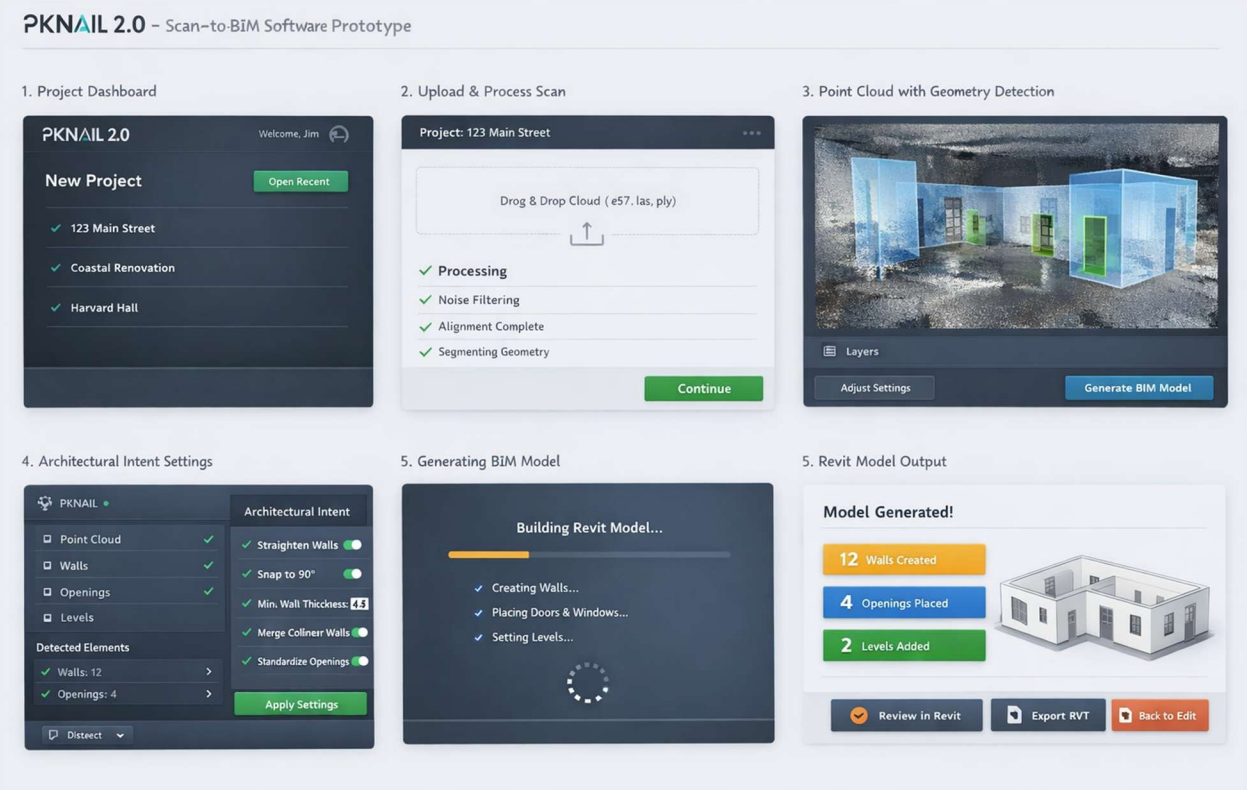
PKNAIL 2.0 — Feasibility & Architecture Brief

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Conceptual Flow:

1. Upload and process scan data
2. Visualize detected geometry (walls, openings, levels)
3. Apply “architectural intent” rules
4. Generate a clean Revit model
5. Review and export

1. Executive Summary

PKNAIL was originally developed as a Revit-integrated engine that converts structured measurement inputs into parametric BIM elements.

We are now exploring whether this engine can be extended to:

Use SLAM-derived point cloud data to assist in generating clean, architecturally correct Revit models.

This is not a full automation effort.

The goal is a guided BIM construction system that combines extracted geometry with rule-based modeling logic.

PKNail 1.0 proved that BIM doesn't need to be drawn — it can be generated from structured field data. PKNail 2.0 simply replaces the human input with point clouds, which puts us significantly closer to a fully automated system than anyone starting from zero."

It is an ARCHITECTURAL INTENT ENGINE

- Snap near-orthogonal walls to 90°
- Join walls within tolerance
- Normalize wall thickness
- Close loops
- Apply dimensional rounding

Anticipated DEVELOPMENT PHASES

Phase 0: Review PKNail 1.0

Phase 1: JSON → Revit pipeline

Phase 2: Simulated geometry

Phase 3: Point cloud processing

Phase 4: Wall generation

Phase 5: Openings

Phase 6: Intent refinement

CORE PRINCIPLES

1. Command layer is central
2. Always validate in Revit
3. Visual debugging required
4. Human-in-loop first

FINAL POSITIONING

PKNAIL 2.0 is a geometry-to-BIM translation engine leveraging a proven rule-based system,enabling scalable, controllable, and high-quality model generation.

2. Core Concept

PKNAIL 1.0

Laser Measurement → Structured Input → PKNAIL → Revit Model

PKNAIL 2.0

Point Cloud → Geometry Extraction → PKNAIL → Revit Model

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3. System Overview

[SLAM Point Cloud]
↓
[Preprocessing] this can be done manually
- **Noise filtering**
- **Drift correction**

↓
[Geometry Extraction] is this possible
- **Walls (planes)**
- **Floors / levels**
- **Openings**

↓
[PKNAIL Engine]
- **Straighten geometry**
- **Apply BIM rules**
- **Place hosted elements**

↓
[Revit Model Output]
↓
[User Review]

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4. What This Is / Is Not

This IS:

- **A controlled BIM generation system**

- **Geometry + rules working together**
- **Focused on usable architectural models**

This is NOT:

- **A fully automated AI scan-to-BIM tool**
- **A mesh reconstruction pipeline**
- **A visualization product**

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5. Key Differentiator — Architectural Intent

PKNAIL does not attempt to reproduce reality exactly.

It constructs a clean, rational model based on architectural intent.

Examples:

- **Crooked wall → straight wall**
- **Noisy scan → simplified geometry**
- **Irregular opening → standardized family**

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6. Existing PKNAIL Capabilities

PKNAIL already supports:

- **Parametric wall creation**
- **Automatic wall joins and intersections**
- **Door and window placement**
- **Hosted element logic**
- **Proper Revit category behavior**

Key Insight:
PKNAIL already encodes how to build BIM correctly.
The opportunity is feeding it from reality.

7. Example Workflow

Input:
Interior SLAM scan

Detected:

- **4 wall planes**
- **1 floor plane**
- **2 openings**

Output:

- **Clean Revit walls**
- **Level created**
- **Parametric openings placed**

User Role:

- **Review**
- **Adjust edge cases**

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8. Phase 1 Definition (Critical)

Initial Focus:

- **Interior SLAM datasets**
- **Orthogonal wall conditions**
- **Limited complexity**

Objective:

- **Generate walls and levels only**

Success Threshold:

- **80%+ wall accuracy**
- **Clean alignment and joins**
- **Model usable without rebuild**

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9. Sample Dataset (Provided)

- **Residential interior scan**
- **2,000–5,000 SF**
- **GeoSLAM Horizon data**

Expected Output:

- **Walls generated within ~1” tolerance**
- **Levels correctly placed**
- **Openings optional (stretch goal)**

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10. Key Technical Questions

Geometry Extraction

- **Reliability of wall plane detection**
- **Handling noise and drift**

Processing Approach

- **Rule-based vs ML-assisted segmentation**

Integration

- **Mapping extracted geometry into PKNAIL**

Revit Output

- **Efficient API strategy**
- **Performance considerations**

11. Known Challenges

- **SLAM drift**
- **Weak planar definition**
- **Vegetation / exterior noise**
- **Opening detection**
- **Wall thickness inference**

12. Success Criteria

- **Reduce modeling time by 50%+**
- **Produce clean, usable Revit models**
- **Require refinement—not full rebuild**

13. Requested Outcome

We are looking for:

1. **Feasibility assessment**
2. **Recommended architecture**
3. **Suggested tech stack**
4. **MVP prototype approach**
5. **Timeline and cost**

14. Materials Provided

- **PKNAIL source code**
- **Sample datasets (if needed)**

15. Closing Thought

PKNAIL solved:

“How do we build BIM correctly from structured inputs?”

PKNAIL 2.0 asks:

“Can we generate those inputs directly from reality/point clouds?”

16. Phase 1 Definition

Initial Focus:

- Interior SLAM datasets
- Orthogonal wall conditions
- Limited complexity

Objective:

- Generate walls and levels only

Success Threshold:

- 80%+ wall accuracy
- Clean alignment and joins
- Model usable without full rebuild

Test Case:

- Single residential dataset (provided)
- Evaluate output vs manual model